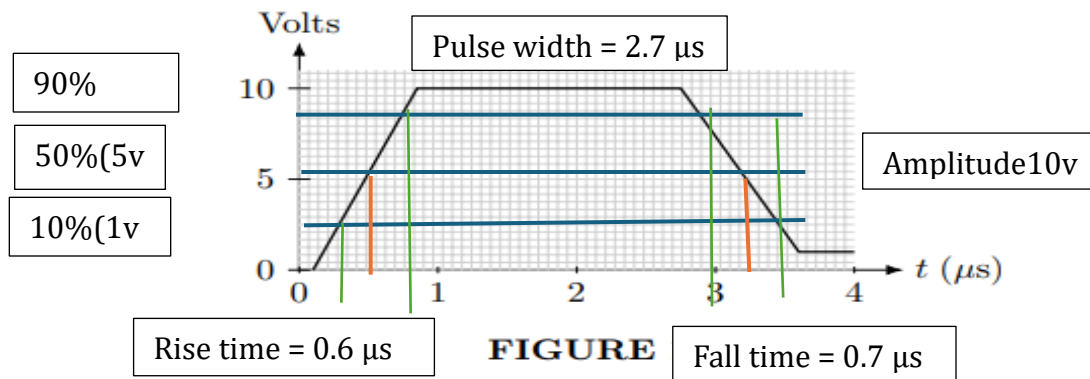


1 Chapter 1

1. For the pulse shown in Figure 1–60, graphically determine the following:

- (a) rise time
- (b) fall time
- (c) pulse width
- (d) amplitude



- a) Rise time: $0.77 - 0.17 = 0.6\mu\text{s}$
- b) Fall time: $3.56 - 2.84 = 0.7\mu\text{s}$
- c) Pulse width: $3.52 - 0.47 = 2.7\mu\text{s}$
- d) Amplitude: 10v

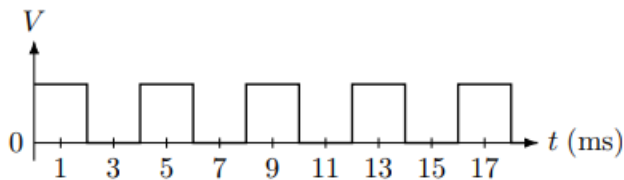


FIGURE 1–61

2. Determine the period of the digital waveform in Figure 1-61

$$T = 4 - 0 = 4\text{ms}$$

3. What is the frequency of the waveform in Figure 1-61

$$F = 1/T$$

$$F = 1/4\text{ms} = 0.25\text{kHz or } 250\text{Hz}$$

4. Is the pulse waveform in Figure 1-61 periodic or non-periodic?

Periodic

5. Determine the duty cycle of the waveform in Figure 1-61

$$\text{Duty} = (\text{Pulse width} / T) * 100\%$$

$$(2\text{ms} / 4\text{ms}) * 100\%$$

$$= 0.25 * 100\% = 50\%$$

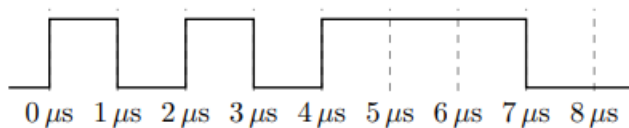


FIGURE 1-62

6. Determine the bit sequence represented by the waveform in Figure 1-62. A bit time is 1 μs in this case

1010110

7. What is the total serial transfer time for the eight bits in Figure 1-62? What is the total parallel transfer time?

$$\text{Serial} = \text{number of bits} * \text{bit time}$$

$$8 * 1 = 8\mu\text{s}$$

$$\text{Parallel} = 1\mu$$

8. What is the period if the clock frequency is 3.5GHz?

$$T = 1 / 3.5 \times 10^9 = 2.857 \times 10^{-10}$$

0.2857ns

9. From a single logical statement from the following information:

- a) The light is ON if SW1 is closed.
- b) The light is ON if SW2 is closed.
- c) The light is OFF if both SW1 and SW2 are open.

The light is ON if SW1 OR SW2 are closed (or both)

10. A logic circuit requires HIGHS on all its inputs to make the output HIGH. What type of logic circuit is it?

AND gate

11. A basic 2- input logic circuit has HIGH on one input and LOW on the other input, and the output is LOW. Identify the circuit.

AND gate

12. A basic 2-input logic circuit has HIGH on one input and a LOW on the other input, and the output HIGH . What type of logic circuit is it?

OR gate

Chapter 2

1. Convert each binary number to decimal:

(a) $110011.11 = 51.375$

(b) $101010.01 = 41.125$

(c) $1000001.111 = 64.4375$

(d) $1111000.101 = 120.3125$

(e) $1011100.10101 = 92.328125$

(f) $1110001.0001 = 113.03125$

2. What is the highest decimal number that can be represented by each of the following numbers of binary digits (bits)?

$2^n - 1$

(a) two = $2^2 - 1 = 3$

(b) three = $2^3 - 1 = 7$

(c) four = $2^4 - 1 = 15$

(d) five = $2^5 - 1 = 31$

(e) six = $2^6 - 1 = 63$

(f) seven = $2^7 - 1 = 127$

(g) eight = $2^8 - 1 = 255$

(h) nine = $2^9 - 1 = 511$

(i) ten = $2^{10} - 1 = 1023$

(j) eleven = $2^{11} - 1 = 2047$

3. Convert each number to binary using repeated division by 2:

(a) 15

$15/2 = 7$, remainder 1, $7/2 = 3$ remainder 1, $3/2 = 1$, remainder 1, $1/2 = 0$ remainder 1 = 1111

(b) 21

$21/2 = 10$ remainder 1, $10/2 = 5$ remainder 0, $5/2 = 2$ remainder 1, $2/2 = 1$, remainder 0, $1/2 = 0$ remainder 1 = 10101

(c) 28

$28/2 = 14$ remainder 0, $14/2 = 7$ remainder 0, $7/2 = 3$ remainder 1, $3/2 = 1$ remainder 1, $1/2 = 0$ remainder 1 = 111000

(d) 34

$34/2 = 17$, remainder 0, $17/2 = 8$ remainder 1, $8/2 = 4$ remainder 0, $4/2 = 2$ remainder 0, $2/2 = 1$ remainder 0, $1/2 = 0$ remainder 1 = 100010

(e) 40

$40/2 = 20$ remainder 0, $20/2 = 10$ remainder 0, $10/2 = 5$ remainder 0, $5/2 = 2$ remainder 1, $2/2 = 1$ remainder 0, $1/2 = 0$ remainder 1 = 101000

(f) 59

$59/2 = 29$, remainder 1, $29/2 = 14$ remainder 1, $14/2 = 7$ remainder 0, $7/2 = 3$ remainder 1, $3/2 = 1$ remainder 1, $1/2 = 0$ remainder 1 = 111011

(g) 65

$65/2 = 32$ remainder 1, $32/2 = 16$ remainder 0, $16/2 = 8$ remainder 0, $8/2 = 4$ remainder 0, $4/2 = 2$ remainder 0, $2/2 = 1$ remainder 0, $1/2 = 0$ remainder 1 = 1000001

(h) 73

$73/2 = 36$ remainder 1, $36/2 = 18$ remainder 0, $18/2 = 9$ remainder 0, $9/2 = 4$ remainder 1, $4/2 = 2$ remainder 0, $2/2 = 1$ remainder 0, $1/2 = 0$ remainder 1 = 10001001

4. Add the binary numbers:

(a) $11 + 01 = 100$

(b) $10 + 10 = 100$

(c) $101 + 11 = 1000$

(d) $111 + 110 = 1101$

(e) $1001 + 101 = 1110$

(f) $1101 + 1011 = 11000$

5. Express each decimal number as an 8-bit number in

the 2's complement form:

$$(a) +12 = 00001100$$

$$(b) -68 = 10111100$$

$$(c) +101 = 01100101$$

$$(d) -125 = 10000001$$

6. Perform each addition in the 2's complement form:

$$(a) 10001100 + 00111001 = 11000101$$

$$(b) 11011001 + 11100111 = 11000000$$